

Conversion Factors

Mass and Density

1 kg	= 2.2046 lb
1 g/cm ³	= 10 ³ kg/m ³
1 g/cm ³	= 62.428 lb/ft ³
1 lb	= 0.4536 kg
1 lb/ft ³	= 0.016018 g/cm ³
1 lb/ft ³	= 16.018 kg/m ³

Length

1 cm	= 0.3937 in.
1 m	= 3.2808 ft
1 in.	= 2.54 cm
1 ft	= 0.3048 m

Velocity

1 km/h	= 0.62137 mile/h
1 mile/h	= 1.6093 km/h

Volume

1 cm ³	= 0.061024 in. ³
1 m ³	= 35.315 ft ³
1 L	= 10 ⁻³ m ³
1 L	= 0.0353 ft ³
1 in. ³	= 16.387 cm ³
1 ft ³	= 0.028317 m ³
1 gal	= 0.13368 ft ³
1 gal	= 3.7854 × 10 ⁻³ m ³

Force

1 N	= 1 kg · m/s ²
1 N	= 0.22481 lbf
1 lbf	= 32.174 lb · ft/s ²
1 lbf	= 4.4482 N

Pressure

1 Pa	= 1 N/m ²
	= 1.4504 × 10 ⁻⁴ lbf/in. ²
1 bar	= 10 ⁵ N/m ²
1 atm	= 1.01325 bar
1 lbf/in. ²	= 6894.8 Pa
1 lbf/in. ²	= 144 lbf/ft ²
1 atm	= 14.696 lbf/in. ²

Energy and Specific Energy

1 J	= 1 N · m = 0.73756 ft · lbf
1 kJ	= 737.56 ft · lbf
1 kJ	= 0.9478 Btu
1 kJ/kg	= 0.42992 Btu/lb
1 ft · lbf	= 1.35582 J
1 Btu	= 778.17 ft · lbf
1 Btu	= 1.0551 kJ
1 Btu/lb	= 2.326 kJ/kg
1 kcal	= 4.1868 kJ

Energy Transfer Rate

1 W	= 1 J/s = 3.413 Btu/h
1 kW	= 1.341 hp
1 Btu/h	= 0.293 W
1 hp	= 2545 Btu/h
1 hp	= 550 ft · lbf/s
1 hp	= 0.7457 kW

Specific Heat

1 kJ/kg · K	= 0.238846 Btu/lb · °R
1 kcal/kg · K	= 1 Btu/lb · °R
1 Btu/lb · °R	= 4.1868 kJ/kg · K

Others

1 ton of refrigeration	= 200 Btu/min = 211 kJ/min
1 volt	= 1 watt per ampere